The potential for abiogenesis within the brine pockets of Titan's ice shell

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July 2026

Abstract

Saturn's largest moon, Titan, holds an organic-rich surface and atmosphere built from complex nitrogen-rich material known as tholins. Far below lies a potential global water ocean, separated by a thick shell of water ice capped with a layer of methane clathrate. This barrier keeps the surface organics and the liquid water apart, so the ingredients for prebiotic chemistry rarely meet. Impact cratering is one of the few natural processes that may be able to bring them together. Pools created by these events may transform into a warm brine pocket inside the shell that persists for hundreds to tens of thousands of years before either freezing or possibly merging with the ocean. This paper attempts to determine whether these pockets could be the hosts of prebiotic chemistry through the use of GFN2 extended tight binding (GFN2-xTB) molecular dynamics (MD) simulations. The hypothetical impactor used for the model is a 1.5 km diameter comet with an initial velocity of 7 $km.s^{-1}$ impacting at 45°. By considering atmospheric deceleration, crater and melt volume relations, and composition of all components, it is possible to estimate the crater size and the molecular inventory within the melt pool contributed by the tholins, the melted methane clathrate cap, and the comet itself. Due to computational constraints, the MD simulations are limited to a few hundred atoms over a few hundred picoseconds. The complex molecular composition must be reduced to a representative mixture of 21 species, consisting of 86 molecules, or 311 atoms, while keeping the species most relevant to prebiotic chemistry. Cometary minerals are disregarded in this study as their relevance is mostly as surface area to catalyse reactions, which would take up a majority of the atoms allocated, and single mineral molecules have little impact on the chemistry of interest. The mixture is optimised in a box, at which point reactive GFN2-xTB MD is run across six independent replicas for 200 ps at 1000 K. This produces trajectories for each atom which are analysed by calculating the Wiberg bond orders (WBO), returning the molecular contents of the box at 50 fs timesteps. Stable reactive chemistry is found to be present across replicas. Within the first hundreds of femtoseconds the trajectories show a radical-rich medium (atomic H^+, thiyl (HS), hydroxyl (OH^-), and formate (HCO_2^-)) while water, methane, ammonia, and aromatic rings persist. New carbon-nitrogen and carbon-sulphur bonds consistently form across replicas, glycine is repeatedly converted into glycyl fragments ($C_2H_4NO_2$), and the cometary-derived phosphorus monoxide radical PO is oxidised and bonded into organics. Large nitrogen-rich clumps of up to ~50 atoms assemble transiently, likely artifacts of the limitations in the analysis. The similar bond-forming steps and heteroatom chemistry present across the replicas indicates that Titan's brine pockets could potentially support the steps to prebiotic chemistry. The elevated 1000 K temperature required for a productive simulation, among other factors such as short timescales, and a hydrogen mass set to 4 amu, mean that these runs act as more of an

indicator of possible chemistry rather than a portrayal of meaningful reaction rates and concentrations.

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1 Introduction

1.1 Titan as a moon of astrobiological interest

For life as we know it to emerge, three components must flawlessly blend in the right environment: the organic building blocks, liquid water as a solvent, and energy to drive reactions. Saturn's moon, Titan, contains an abundance of the former two, separated by ~100 km of ice [CITE]. It is the only moon in the Solar System with a substantial atmosphere, with surface pressures of ~1.5 atm, and the only planetary body besides the Earth known to hold stable liquid on its surface[1], though its lakes, rivers, and seas instead of being composed of water are a mixture of liquid methane and ethane. Its thick nitrogen and methane atmosphere allows for complex photochemistry in its upper atmosphere, driven mostly by Saturn's magnetospheric particles, as well as cosmic rays and solar UV. This process produces a particular classification of nitrile-rich polycyclic aromatic hydrocarbons known as tholins, which settle onto the surface and build up as thick layers of complex organic material[2,3]. Beneath the outer shell of water ice, gravity and electric-field measurements from the Cassini-Huygens mission suggest that a global subsurface ocean of liquid water may be present, which could hold high concentrations of dissolved ammonia acting as an antifreeze[4,5]. The existence of this ocean has never been confirmed directly[4] and a recent paper from JPL [CITE] suggests that it may be more of a dynamic 'slush' rather than a stable liquid environment.

1.2 Impact-generated melt as a habitable environment

If the ocean is assumed to be present, in order for water-based life to have the potential to emerge, the surface tholins and subsurface ocean must react. Impact cratering may be one of the few, if not the only, natural process that could bridge this gap. When a comet or asteroid strikes Titan's shell, some of its kinetic energy is deposited as heat, melting the ice and forming a pocket of liquid water that can last hundreds to tens of thousands of years depending on its size and depth[6,7], freezing over the top and slowly migrating downwards as a self-contained reaction vessel. The methane clathrate that caps the shell has strong insulatory properties, lengthening the possible lifespan of the liquid solution. These pockets would hold the surface organics, the methane-clathrate material of the cap, and the organic, volatile, and mineral content of the impactor itself. In this environment, reactions can proceed in a smaller and more concentrated setting before merging into the highly dilute ocean, assuming that it exists and the pocket reaches it before refreezing. This pocket resembles the impact-driven environments that some believe to have aided in the origin of life on the early Earth[8].

Cometary impactors are of particular interest, and they have been shown to contain various volatiles, minerals, and organics relevant to abiogenesis, including the simplest amino acid, glycine, which has been found in material returned by the Stardust mission[9,10]. The melted clathrate cap adds further water and methane, and the tholins add nitrogen-bearing aromatics and nitriles. The result is a thick, warm brine stocked with prebiotic feedstock and all the elements thought to be necessary for life: Carbon, hydrogen, nitrogen, oxygen, phosphorus, and sulfur (CHNOPS).

1.3 Prebiotic pathways

Several of the molecules expected to be present in these pockets are thought to be central to prebiotic chemistry, and laboratory work over the last century has shown the potential of what they can build. Through cyanosulfidic chemistry, HCN, present in high concentrations in tholins, and related species give rise to precursors of nucleotides, amino acids and lipids necessary for all known lifeforms[12,13]. Formamide (CH_3NO), present in comets, offers a more direct route to nucleobases and forms from HCN and water[14]. Formaldehyde (CH_2O), also present in comets, is a key part of the formose reaction which builds sugars, and neighboring ammonia (NH_3) supplies nitrogen for amination reactions. The legendary Miller-Urey experiment demonstrated that amino acids and other building blocks form spontaneously from similar simple precursors when an energy source is present[15]. For Titan specifically, laboratory hydrolysis of tholins has been shown to produce prebiotically relevant molecules, including amino acids, indicating that Titan’s own surface organics are a potential source of prebiotic compounds once they interact with liquid water[16].

1.4 Reactive molecular dynamics simulations

Laboratory experiments cannot directly observe the earliest sub-picosecond bond-forming reactions that occur, and they cannot easily separate the contribution of each precursor in a complex solution such as this one. Reactive MD simulations would complement these experiments by following bond breaking and formation occurrences step by step. MD has reproduced Miller-Urey-like reactions from first principles[17], and MD nanoreactors have simulated known prebiotic reaction networks from simple feedstocks[18]. These computational methods, however, are expensive for solutions of thousands of atoms over long timescales. The semi-empirical tight-binding method, GFN2-xTB, used in this paper reproduces accurate quantum mechanical depictions of bonding and charge for 311 atoms over 200 picoseconds[19,20], and is therefore well suited to testing whether a Titan brine pocket can produce prebiotically relevant chemistry.

1.5 Aims

This study follows the composition and early chemistry of a modelled hypothetical Titan brine pocket and attempts to address three main questions: What is a plausible molecular inventory of a brine pocket formed by a cometary impact? When a computationally allowing analogue of that pocket is simulated with GFN2-xTB MD, can reactions indicative of prebiotic chemistry occur spontaneously? Which feedstock species contribute most to the production of interesting products, and what does this imply for the habitability and potential for abiogenesis within the brine pockets? Section 2 describes the impact and composition model as well as the MD simulation and analysis protocol. Section 3 reports the molecular inventory and trajectories of the simulations. Section 4 interprets these results in the context of Titan's habitability and outlines the limitations of the method. Section 5 summarises the prebiotically interesting chemistry found in this paper and as well as the importance of further in-silica, in-vitro, and in-situ research.

2 Methods

Three main phases contribute to the results of this paper, summarised in Figure 1. A hypothetical cometary impact model with specific parameters determines the molecular inventory of a brine pocket (Sections 2.1 and 2.2). An adjusted, but representative within the limitations of the simulation, subset of that composition is used as the input for a GFN2-xTB MD simulation[20,21] (Sections 2.3 to 2.6), after which the trajectories are analysed for molecular products (Sections 2.7 and 2.8).

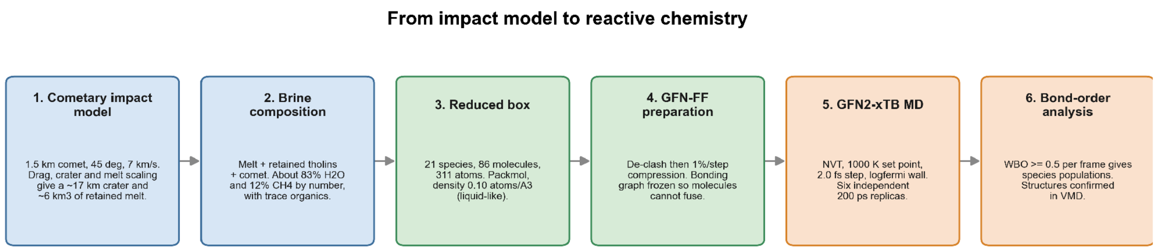

Figure 1. ***[Unsure, might remove, definitely will change]*** *Workflow, from the cometary impact model that determines the brine composition (Sections 2.1 and 2.2), to the reduced simulation box (Sections 2.3 and 2.4) and its GFN-FF preparation (Section 2.5), to the GFN2-xTB MD simulation (Section 2.6) and the WBO-backed trajectory analysis (Sections 2.7 and 2.8).*

2.1 Impact and brine pocket formation model

The hypothetical impactor used to estimate brine pocket composition is a comet of diameter D = 1.5 km and density ρ = 600 kg.m^{-3}, entering the top of Titan's atmosphere with velocity v_0 = 7 km.s^{-1} at an impact angle θ = 45°. These parameters were chosen as they correspond to the smallest comet that suffers negligible atmospheric mass loss and ablation while still producing a concentrated brine. The deceleration from the atmosphere is calculated from an exponential drag relation, which gives a surface velocity

$$v_{surf} = v_0 \exp[-3C_d\Sigma / (4D\rho\sin\theta)],$$

where $C_d = 1$ is the drag coefficient for a hypersonic sphere and $\Sigma \approx 1.09 \times 10^5$ kg.m^{-2} is the atmospheric column mass P_{surf}/g_T (surface pressure divided by Titan's surface gravity). This leads to the comet reaching the surface at $v_{surf} \approx 6.3$ km.s^{-1}. The crater diameter immediately after impact is taken from the methane clathrate cap scaling law of Wakita et al.[22], where for a 5 km thick cap and an impactor diameter $D_{imp} < 3.536$ km, then

$$\log(D_{crater}) = 0.918 \log(D_{imp} + 1.225),$$

assuming a 10.5 km.s^{-1} impactor velocity. This value is then corrected for the desired velocity v_{surf} following scaling laws from Johnson et al.[23], where $D_{crater} \propto v_{surf}^{0.5}$, and for impactor density using π-group scaling[24,25] with $D_{crater} \propto \rho^{0.3}$. This gives a crater ~16.6 km in diameter.

The volume of impact melt is estimated from a melt-volume scaling law and initial V_{melt} values derived by Kalousová et al.[26], $V_{melt} \propto \rho D^3 v_{surf}^{1.3}$, with the v_{surf} exponent of 1.3 taken from Pierazzo et al.[27] for ice-ice impacts, producing a value of $V_{melt} \approx 6.6$ km^3. The fraction of melt that remains in the crater, rather than being vaporised or ejected, is scaled from the Titan impact melt results of the modelling by Artemieva and Lunine[6], where retained volume $V_{melt,ret} \propto \rho D^3 v_{surf}^{1.9}$. The exponent of 1.9 accounts for the 460 kg.m^{-3} density that Artemieva and Lunine[6] assumes for their entirely porous ice comet, with the 1.3 from earlier only being pertinent to overall melt volume. This process leaves us with ~6.3 km^3 of retained melt from the ice sheet. This melt is assumed to come entirely from the methane-clathrate cap, although the crater may actually reach slightly below this layer, and its composition is inferred using the standard clathrate composition of $CH_4.5.75H_2O$ [CITE: https://doi.org/10.1029/2002JB001872], corresponding to a total retained melt mass of $M_{melt,ret} \approx 5.3 \times 10^{12}$ kg.

The mass of the comet is calculated to be ~1.1×10^{12} kg, and using a value of 0.5 for the fraction of cometary mass retained in the crater derived by Artemieva and Pierazzo [CITE: https://doi.org/10.1111/j.1945-5100.2011.01195.x], this leaves us with a retained cometary mass of $M_{comet,ret} \approx 5.3 \times 10^{11}$ kg.

The tholins are assumed to have been deposited in a 10 m thick layer on the surface of the ice sheet (true value between 4 and 30 m [CITE https://doi.org/10.1006/icar.1994.1191]) with a density of 1500 kg.m^{-3} [CITE: https://doi.org/10.1016/j.icarus.2005.11.012]. The fractions of these retained in the crater are assumed to be 10% of the surface layer volume covering the crater area[7], leading to a final mass of retained tholins of $M_{tholins,ret} \approx 3.2 \times 10^{11}$ kg.

2.2 Composition of the brine pocket

The molecular composition of the pocket contains compounds from three sources: The surface melt, the tholins, and the cometary material (split into ice, organics, and minerals). The retained melt is the most prominent source of material in terms of mass, and its composition is discussed above. The retained tholins, having an incredibly complex composition, are simplified by accounting only for the seven most prominent functional groups [CITE: Imanaka 2004] (being nitriles, isonitriles, aliphatic hydrocarbons, amines, pyrroles, pyridines, and imines) which are assumed to have equal molecular abundance, with the exception of isonitriles which are ~10% as abundant as nitriles [CITE]. The retained cometary mass is split into ice, organics, and minerals in mass proportions of 2:1:1, respectively [CITE]. Using spectral data from comet 67P/Churyumov–Gerasimenko [CITE: Rubin et al., Altwegg et al.] and 9P/Tempel 1 [CITE: Lisse et al.], the relative abundances of each cometary material are found and the most prominent molecules are considered in each case. Due to limitations in the reactive MD software, all species, with the exception of those specifically pertinent to prebiotic chemistry, are simplified down to the simplest molecules that contain the functional groups of the original compound (Appendix 1). The mass of each source is converted to a molecule count using the species' molar masses, and the counts are normalised over the total mass of brine in the crater. The resulting solution is dominated by water at ~83% of all molecules, followed by methane at ~12%, and the rest split among volatiles, nitriles, aromatics, minerals, and organics (Figure 2). The mixing ratios found inform the starting composition for the MD simulations.

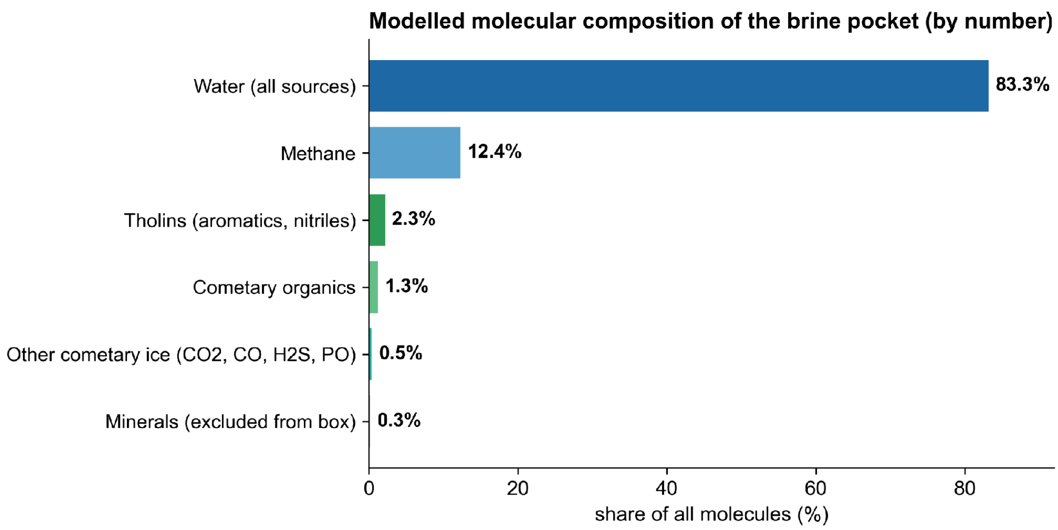

Figure 2. ***[Maybe replace this with Appendix 1 or table derived from it]*** *Modelled molecular composition of the brine pocket by number of molecules. Includes retained clathrate melt, cometary ice, tholins, and cometary organics. Water and methane dominate 95.7% of the molecular inventory.*

2.3 Simplification of the brine composition for xTB simulations

Given computational limitations, GFN2-xTB MD is only practical for a few hundred atoms over a few hundred picoseconds, meaning the full inventory discussed above is adjusted to a box containing 21 species, composed of 86 molecules or 311 atoms. This process keeps water and methane as the dominant species, and includes at least one molecule that is simplified from every prebiotically important class in the inventory. These are the nitriles (HCN), an isonitrile (HNC), an amino acid (glycine), an amide (formamide), an isocyanate (isocyanic acid), aldehydes (formaldehyde and acetaldehyde), an imine (methanimine), an alcohol (methanol), a carboxylic acid (formic acid), nitrogen-containing aromatics (pyridine and pyrrole), an amine (methylamine), a thiocarbonyl (thioformaldehyde), a phosphorus carrier (PO), and the volatiles CO_2, CO, NH_3 and H_2S. In order to showcase the chemistry with these compounds in a box of only a few hundred atoms, many trace organics are over-represented, especially glycine which is natively present in comets at about 170 parts per billion [CITE: PMID: 33798393]. The cometary minerals are excluded in entirety, as they would need to be composed of ~70% of the allocated atoms to form a mineral surface in which reactions can be catalysed, and the chemistry of interest would make little use of free floating mineral molecules such as forsterite, ferrosilite, or olivine which realistically would be present. The final simulation box composition is given in Table 1.

Table 1. *Composition of the simulated brine pocket (86 molecules, 311 atoms), with the origin of each species.*

Molecule	Formula	N	Origin
Water	H_2O	60	Melt and cometary ice
Methane	CH_4	5	Melt and cometary ice
Pyridine	C_5H_5N	1	Tholins
Pyrrole	C_4H_5N	1	Tholins
Methylamine	CH_5N	1	Tholins
Hydrogen cyanide	HCN	2	Tholins and cometary organics
Hydrogen isocyanide	HNC	1	Tholins and cometary organics
Glycine	$C_2H_5NO_2$	1	Cometary organics
Formamide	CH_3NO	1	Cometary organics
Methanimine	CH_3N	1	Cometary organics
Acetaldehyde	C_2H_4O	1	Cometary organics
Methanol	CH_4O	1	Cometary organics
Formaldehyde	CH_2O	2	Cometary organics
Formic acid	CH_2O_2	1	Cometary organics
Isocyanic acid	HNCO	1	Cometary organics
Ammonia	NH_3	1	Cometary ice
Carbon dioxide	CO_2	1	Cometary ice
Carbon monoxide	CO	1	Cometary ice
Hydrogen sulphide	H_2S	1	Cometary ice
Thioformaldehyde	CH_2S	1	Cometary organics
Phosphorus monoxide	PO	1	Cometary ice

2.4 Molecule creation and box assembly

Each distinct molecule was built in IQmol [CITE] and optimised with GFN2-xTB at the extreme convergence level, producing accurate geometries. Six independent boxes were then assembled with Packmol[29], using an inter-molecular tolerance of 2.0 Å and a randomised seed for each box, creating geometrically distinct replicas. The molecules were initially packed at a low density of 0.04 atoms.$Å^{-3}$, into a sphere of radius 11.3 Å, slightly smaller than the wall used for the following geometry optimisation of the entire solution so that the packing has room to relax before it is compressed into the desired density of 0.10 atoms.$Å^{-3}$.

2.5 Geometry preparation

Each box was optimised with GFN-FF (GFN - force field)[30], which acts purely as a classical geometry optimiser and does not allow for bond formation. GFN2, as used in the MD runs, takes into account quantum effects in order to determine the lowest energy state, which would rapidly collapse the molecules into large aggregates during an optimisation run. To begin, a 'crude' level optimisation with GFN-FF to simply remove atomic overlaps was run inside a logfermi spherical wall of radius 12.29 Å. The geometry and wall are then procedurally compressed and reoptimised with 'loose' level GFN-FF with steps of 1% of wall radius until it reaches 9.06 Å, producing the desired density of 0.10 atoms.$Å^{-3}$, similar to that of liquid water. Throughout the optimisations the per-atom displacement was capped at 0.05 Å and the lowest Hessian eigenvalue was floored at 0.01 to keep the relaxation stable. A Wiberg bond order (WBO) check at the end was performed to confirm that each box remained a set of separate molecules identical to those initially input into the system.

2.6 Molecular dynamics

The reactive MD run used GFN2-xTB in the NVT ensemble for six independent replicas and was allowed to run for 200 ps. The Berendsen thermostat is set to 1000 K. Although the realistic temperatures would be ~176 - 273 K, the higher value was chosen so that reaction barriers could be crossed within the short simulation time[17,18] [CITE 176-273K]. The integration time step was 2.0 fs, allowed for by altering hydrogen mass to 4 amu. This allowed the simulation time to be increased four-fold at the cost of slowing the fast X-H stretching motions. This repartitioning slowed hydrogen-transfer rates, meaning the outcome of these runs should be considered as a screen for possible chemistry rather than a measurement of realistic rates and compositions. SHAKE constraints were disabled so that bonds involving hydrogen are free to break and form, and the coordinates of each atom were saved every 50 fs. The electronic temperature is set at 300 K to stop overdelocalisation of electron density within the orbitals, which if too high can cause rapid unrealistic formation of covalent bonds. The self-consistent-charge accuracy (SCCACC) was set to 1.5, where 1 is realistic and required with metals present, and 2 is the default setting. The 1.5 value allows for decent force quality while not overly slowing down the simulation. A logfermi wall with steepness $\beta = 6$ and Fermi temperature 3000 K is adopted to keep the molecules from escaping the box and ensures the density remains stable. The number of unpaired electrons (UHF) is set to 1 to account for the single PO radical.

2.7 Trajectory analysis

Molecular species were identified from each saved 50 fs frame with a Python script. The presence of each bond was calculated using WBO, with a bond assigned wherever the bond order was at least 0.5. Double and triple bonds were determined by WBO of 1.5-2.5 and 2.5-3.5, respectively. This method was used instead of a simple distance cutoff since, at liquid density, a covalent radius cutoff[31] merges the whole box into one connected aggregate of atoms. Each connected group of atoms was counted as a molecule and labelled by its chemical formula and bonding pattern so that isomers can be told apart. A population time series was then produced for each species in every replica. Transient molecules, those only present for a couple of frames, are largely disregarded as artifacts, and the rest were analysed by how persistent they were and by how many of the six replicas they were produced in. A WBO threshold of 0.5 can at times consider strong hydrogen bonds as covalent, so the larger molecules should be considered to be independent clustered compounds.

2.8 Visualisation of the trajectories

The trajectories were further inspected using Visual Molecular Dynamics (VMD 1.9.2)[32] in order to get a better view of specific interesting bonding events and to confirm that the products reported by the WBO analysis matched real structures.

3 Results

3.1 Trajectory stability and sampled temperature

The reactive MD simulations ran with the self-consistent charge cycle, which calculates electrostatic potential and charges, successfully converging at every step. All atoms remained within the boundaries set by the logfermi wall throughout every run. Due to the exothermic nature of bond formation, and since the Berendsen thermostat has a finite coupling time, the temperatures calculated during the runs were consistently far above the 1000 K given to the software, with time averages between 1719 and 1898 K (Table 2). The overall mean across the six replicas is ~1828 K. The Berendsen thermostat is however infamously inaccurate for thermodynamical analysis and these values should not be used for any chemical analysis. [IN 200ps RUN IT MAY BRING IT BACK DOWN TO 1000. IF SO, REWRITE]. The largest clusters of atoms formed through the limitations of WBO analysis range from 22 to 49 atoms. These, unless stated otherwise in the following sections, should be considered artifacts rather than true formed molecules (Table 2).

Table 2. Summary of Berendsen thermostat values, and size and composition of largest cluster of atoms in each of the six replicas.

Replica	Time-averaged temperature ⟨T⟩ [K]	Largest cluster [atoms]	Composition of largest cluster
1	1821 (1754–1885)	22	$C_6H_{10}NO_3PS$
2	1898 (1773–1993)	49	$C_{14}H_{24}N_5O_6$
3	1803 (1737–1862)	25	$C_6H_{10}NO_7P$
4	1719 (1646–1783)	26	$C_8H_{14}N_2O_2$
5	1859 (1785–1926)	26	$C_8H_{15}N_2O$
6	1868 (1728–1956)	23	$C_7H_{13}N_2S$

3.2 Evolution of molecular populations

Within the first frames, the solution began shedding hydrogen atoms and small radicals such as atomic H^+, hydroxyl (OH^-), thiyl (HS), and formate (CHO_2^-), with the former two originating in large part from water. Overall, water, methane, ammonia, and pyrrole were present in nearly every frame of each replica, while pyridine, methanol, HCN, and HNC eventually reacted with other compounds [Just wrong, will prob be very different once final data analysed]. The number of distinct product species climbed within the first 500 fs and then fluctuated around a steady level (15?), and the largest bonded cluster reached transient maxima of 22 to 49 atoms across the six replicas (Figure 4) (I think I should remove the cluster from the figure). Each replica demonstrates similar species persisting, being consumed, and being produced (Figure 5). The water and methane solvent remain across the entire run, while the organic feedstock is consumed, and the products begin accumulating and reacting soon after beginning the simulation.

Figure 3. *VMD SCREENSHOTS?*

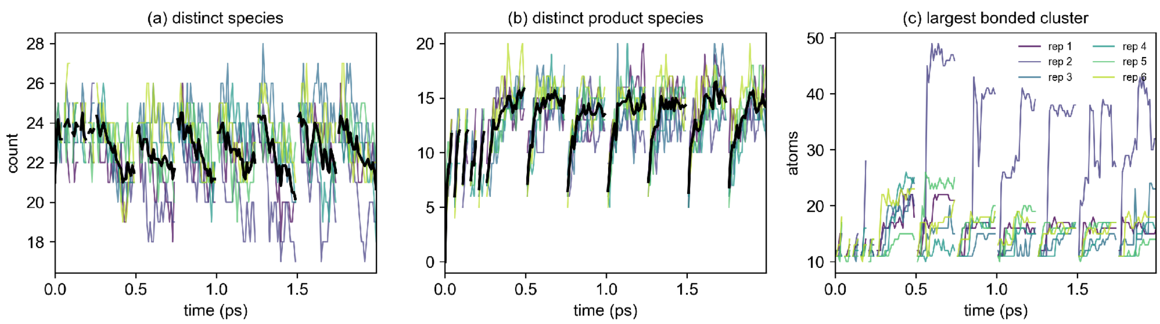

Figure 4. *[Weird periodicity is from glitch in my code. Has been fixed] Molecular populations across the six replicas, with the mean depicted in black. [Add legends to distinguish replicas]* ***(a)*** *The number of distinct chemical species over time. Slightly decreases but remains near original value as fragments form and recombine.* ***(b)*** *The number of distinct product species over time. Rises over the first 0.1 ps and then holds relatively steady.*

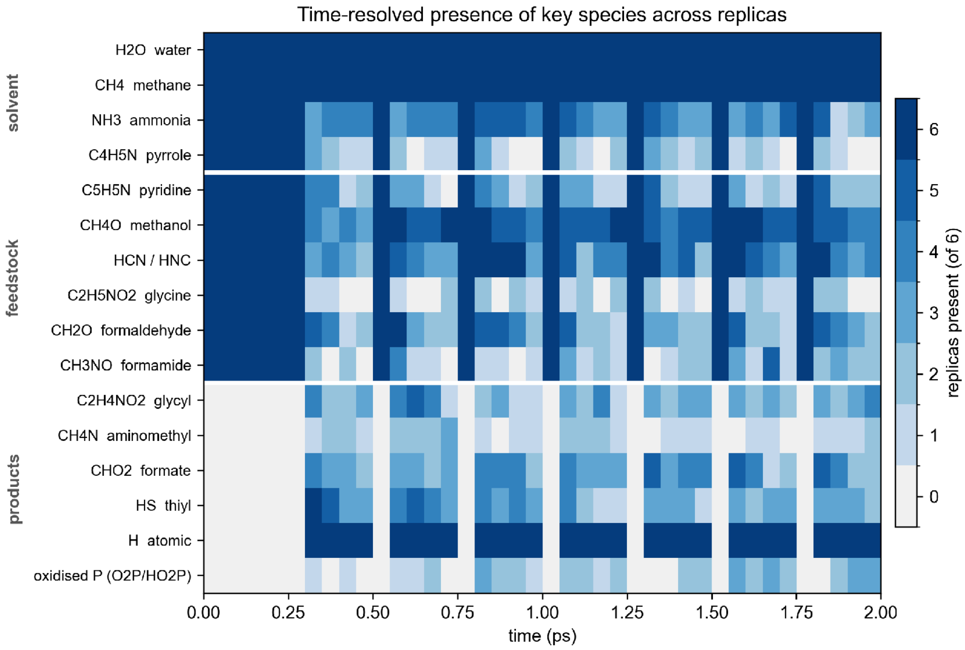

Figure 5. *Time-resolved presence of the key species across the six replicas, grouped into solvent, feedstock, and products. [Expand to include all starting molecules and non-transient products?]. Each cell is shaded by the number of replicas in which that species is present at that timepoint.*

3.3 Products that recur across replicas

The strongest evidence for reproducible chemistry comes from products that appear in all six boxes, since these are not an anomaly brought on by one particular starting geometry. Several intriguing C-N and C-O bonds consistently recur in this manner (Table 3 and Figure 6). A glycine-derived amino-acid fragment, glycyl ($C_2H_4NO_2$) formed in all six replicas, as did the aminomethyl radical (CH_4N) and the nitrogen carriers CN, CNO and CH_2NO. Formate (CHO_2) and thiyl (HS) also appeared in every replica. This consistency shows that C-N bond formation, the mobilisation of glycine, and the release of sulphur from H_2S and thioformaldehyde (H_2CS) are direct results of brine pocket chemistry. [If larger molecules are consistent in real runs discuss these].

The phosphorus in the PO radical unsurprisingly reacted in every box, although the products differed across replicas. The PO was oxidised to phosphite (HPO_3^{2-}) and phosphate (PO_4^{3-}) ions, [TALK ABOUT THE MOVEMENT OF THE P IN EACH REPLICA]. Sulphur behaved similarly, consistently forming thiyl radicals from H_2S and thioformaldehyde [How did it react with organics in each replica].

Table 3. Products that recur across replicas and their prebiotic relevance [EXPAND].

Molecule	**Number of replicas present**	**Prebiotic relevance**
$C_2H_4NO_2$ (glycyl fragment)	6	Glycine-derived amino-acid fragment, sign of mobilised glycine.
CH_4N (aminomethyl)	6	Reactive C-N carrier for organic chain growth
CN, CNO, CH_2NO	6	New C-N and C-N-O species
CHO_2 (formate)	6	Prebiotic building block
HS (thiyl)	6	Allows for sulphur redox
Atomic hydrogen (H^+)	6	Highly reactive and can catalyse organic chemistry
Oxidised phosphorus ($PO4^{3-}$, HPO_3^{2-})	5	Phosphite and phosphate, essential for life and drives reactions
Organophosphorus (C-P bonds)	5	Phosphorus bonded into organic products, essential for life

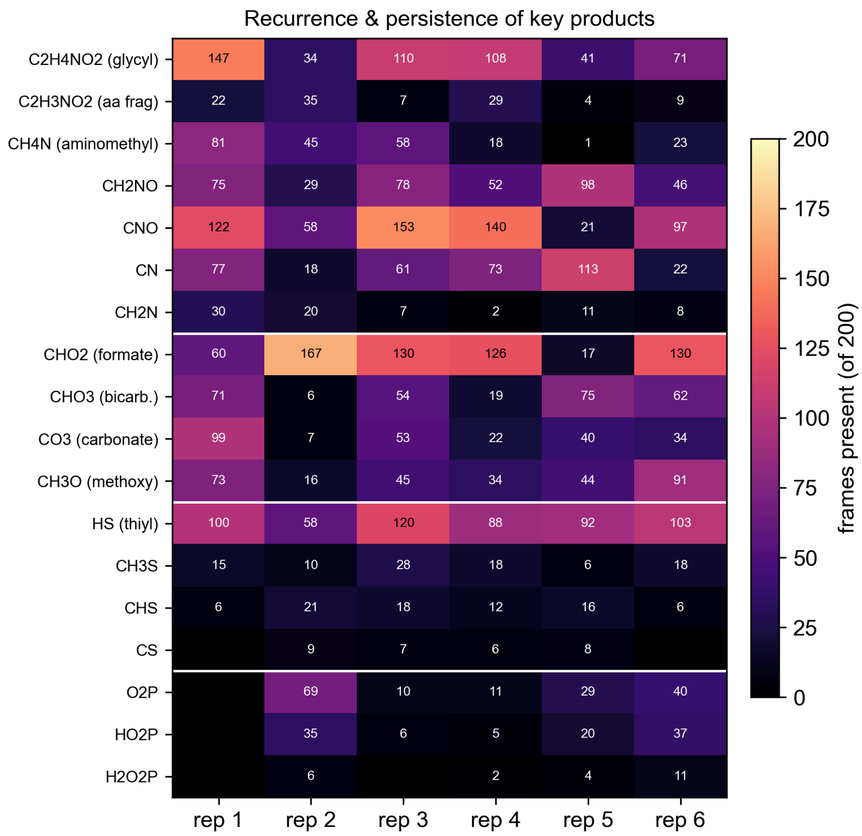

Figure 6. *Recurrence and persistence of key products across the six replicas [Expand]. Each cell shows how many of the 50 fs frames of each box it is present in, from black (none) to bright (4000).*

3.4 Prebiotically significant products

The simulations all formed new carbon-nitrogen, carbon-sulphur, and carbon-phosphorus bonds from the simple feedstock initially provided, with C-N formation being the most common by far, and C-S and C-P products appearing rarely but consistently due to the low initial elemental S and P concentrations (Figure 7). Glycine repeatedly lost and regained hydrogen, appearing as the glycyl fragment in every replica, and was drawn into larger nitrogen-bearing products [Talk about which and consistency once I have final data]. HCN and HNC were mostly consumed as they are highly reactive C-N donors.

[… more cool products that I find]

The simulated Titan brine, enriched with trace organics and volatiles, forms new C-N bonds, mobilises amino acids, and rapidly converts phosphorus and sulphur into reactive species of immense interest to prebiotic chemistry (Figure 8).

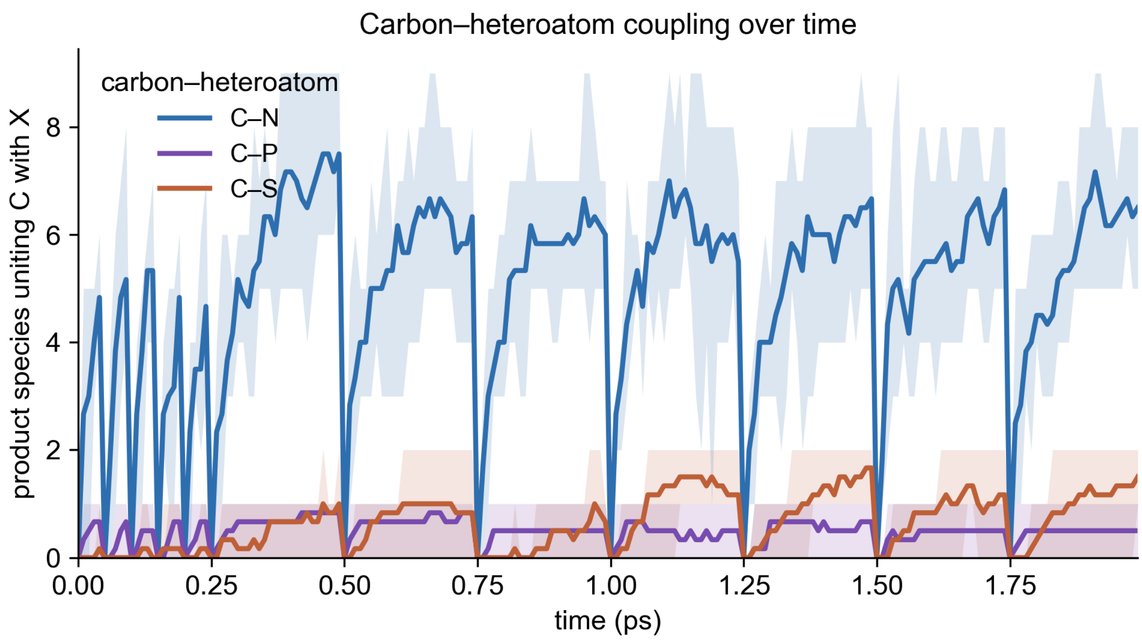

Figure 7. *Carbon–heteroatom coupling over time. For every 50 fs timestep frame, the number of product species that contain the carbon-heteroatoms C-N, C-P, and C-S [Include C-O??]. The bold line is the mean over the six replicas and the shaded band represents their range. C-N bonds are the most common, while C-S and C-P containing products are more scarce but still present in every simulation.*

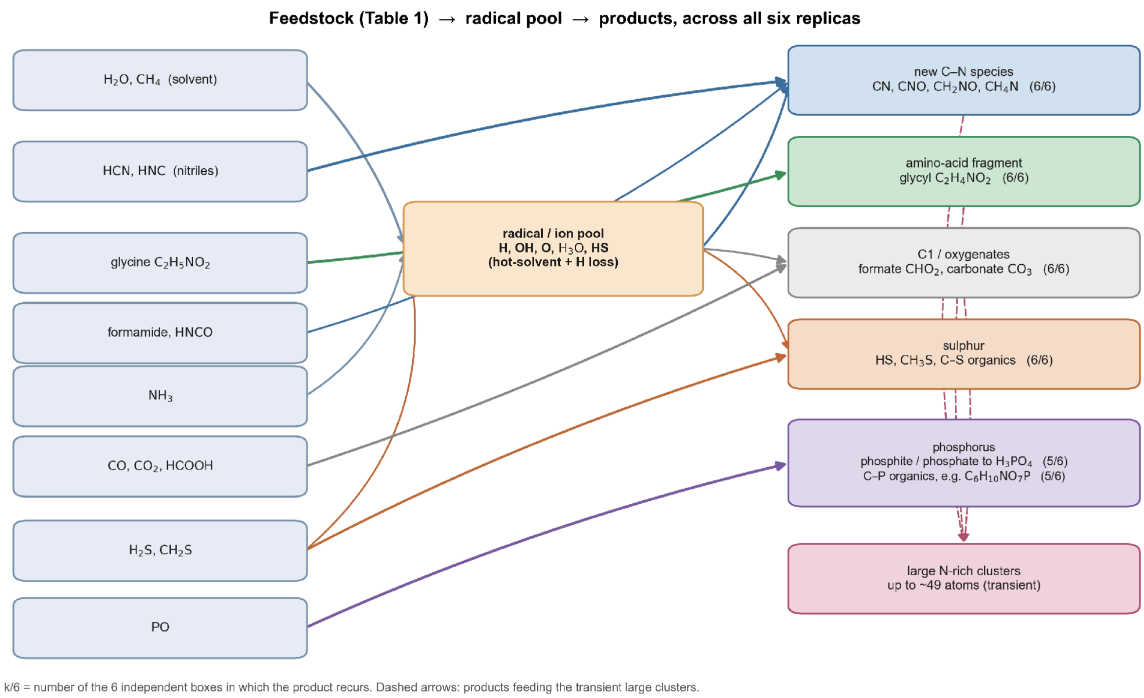

Figure 8. ***[NOT SURE IF I LIKE, MIGHT REMOVE]*** *Reaction network present across replicas.*

4 Discussion

4.1 What Titan's brine pocket chemistry implies

The simulations demonstrate that one of the most important, and in many cases most difficult, steps of prebiotic chemistry is the formation of new C-N bonds. This happens spontaneously in a simulated Titan brine. The recurrence of mobilised glycine, as the glycyl fragment, in all six replicas shows that glycine does not remain inert, but reacts, having potentially huge implications if other amino acids are present. HCN and HNC are the main reactive C-N donors, and would be commonplace in true Titan conditions. Their consumption is consistent with the known reaction pathways leading to amino acids and nucleobases[11,12], while formamide and ammonia provide the amide and amination chemistry complementing the formation of nucleobases[14]. The recurring oxidation of phosphorus monoxide to phosphate and its incorporation into organics in every run of the simulation is a potentially exciting result, as phosphorylation can be a hard step to achieve abiotically. However, it must be taken into account that the concentrations of phosphorus in true brine pocket conditions would be orders of magnitude lower, with a presence of about 0.00011 moles of PO per mole of water in cometary ice [cite: Rubin 2023]. [Actual results reveal anything else?]

4.2 Role of temperature, timescale and system size

The runs sat well above the 1000 K set point, at sampled averages near 1800 K [Will be different!!], because exothermic bond formation overran the Berendsen thermostat's ability to bring the solution towards baseline. This is far above any realistic Titan brine pocket, which would likely be near the water-ammonia eutectic, ~176 K to 273 K [CITE]. The high temperature is used as a kinetic accelerant in order to bring reactions into the time windows that the simulation can observe. The chemistry is therefore biased toward fast, thermally accessible pathways, and at this temperature the water solvent reversibly dissociates. A real pocket would reach the same chemistry, but much more slowly over its lifespan of hundreds to tens of thousands of years[6,7]. The hydrogen mass repartitioning discussed in the methods that allows the 2 fs time step slows hydrogen-transfer rates, reinforcing the need to view the results as chemistry that is possible in these conditions rather than using it as a proxy for realistic reaction rates and equilibrium concentrations. The runs are also small, only 311 atoms, which samples a tiny volume of the estimated 6.2 x 10^{12} kg pool. The absence of any given product or reaction pathway therefore does not indicate that these conditions do not allow for it, rather that further experiments are needed to determine if these reactions are plausible in Titan's conditions.

4.3 Why minerals were excluded

The cometary minerals were left out because their catalytic uses arise when they are present as clusters with large surfaces, where a few dissolved molecules cannot represent the true importance of these compounds to prebiotic chemistry. To overcome this, a slab model should be the next step, where a solid cometary-relevant mineral surface within the simulated box, composed of forsterite, ferrosilite, Fe and Mg niningerite, and amorphous olivine [Cite: Lisse 2006] could both concentrate the organics and lower the reaction barriers to allow for a more accurate chemistry to be represented.

4.3 Heteroatom chemistry

Despite the absence of these minerals, the heteroatom chemistry that did occur is encouraging. [TALK ABOUT NITROGEN MOVEMENT AND MAYBE OXYGEN?] Phosphorus was oxydised into phosphite and phosphate, and was bonded into organics in five of the six boxes [Might change, and mention exact formed products]. Sulphur from H_2S and thioformaldehyde converted into thiyl radicals in every replica and eventually into larger carbon-sulphur species [Again, mention which]. These elements are necessary to all known life, and can be bottlenecks to prebiotic chemistry. Their spontaneous reactions, demonstrated in these simulations, indicate their pathways deserve to be analysed further in future studies.

4.4 Comparison with previous work and implications

These results should be analysed through the lens of similar reactive-MD studies of the origin of life centered brine chemistry that produced prebiotically relevant products and reaction networks from simple feedstocks[17,18]. This paper extends that approach to admire a specific mysterious environment on Titan, brought on by a cometary impactor, and reveals the presence of similar chemistry. There is work on the alteration of Titan' s surface organics and habitability of impact melt done by Artemieva and Lunine[6,7], whose models largely drove the determination of the initial brine compositions used here, though no reactive MD simulations have been published regarding these specific environments. Artemieva and Lunine[6,7], as well as Neish et al.[16] through in-vitro work, discussed tholin hydrolysis as a source of amino acids, and also showed that similar feedstocks form new C-N bonds and mobilise amino acids. This study is attempting to push the importance of impact-generated structures as the locations of promising chemistry, especially on Titan where its surface is natively uninhabitable by life as we know it. The Dragonfly mission, landing in the Shangri-La dune fields of Titan in 2034, will carry the DraMS gas chromatography mass spectrometer[33]. This instrument is built to sample Tholins and once it arrives at Selk crater it will be in the perfect position to analyse the soil altered by cratering events, hopefully driving further in-silica and in-vitro work, with increasingly accurate parameters.

4.5 Limitations and future work

The main caveats to the type of simulation used in this study are computational constraints, leading to an elevated temperature, a small system, and short timescales, along with hydrogen-mass repartitioning as discussed above. These factors alter the chemistry that is accessible, and further studies with more compute and advanced modelling software, such as tblite which is currently in development [CITE], will be able to produce vast quantities of data with fewer limitations. The use of a WBO cutoff of 0.5 counts strong hydrogen bonds as covalent ones, indicating that the largest clusters are likely amalgamations of independent molecules. Important next steps in this research include a temperature ladder, for example 200 K, 400 K and 600 K, to extract activation trends, then longer trajectories or enhanced sampling such as metadynamics to unearth rare reactions, an explicit mineral slab model, and larger boxes. Tblite could support all of these once it is past its experimental phase [CITE AGAIN]. The chosen comet, while convenient for observing chemistry on a 200 ps timescale, probably would not produce a pocket large or hot enough to reach the supposed subsurface ocean, and would freeze relatively rapidly, so that any prebiotic chemistry occurring in there would ultimately have no impact on habitability. Here the possible brine pocket chemistry is in question, and once we have gained a deep understanding about the mechanics in Titan's ice sheet and deep interior we could determine the optimal impact that would be able to produce a brine pocket that would persist over long timescales while simultaneously having optimal chemistry that could allow for abiogenesis.

[RESTATE CONCENTRATION INACCURACIES, & SIMPLIFICATION OF COMPLEX MOLECULES INTO SIMPLEST FORMS WITH THOSE FUNCTIONAL GROUPS.

ALSO IN REAL POCKET METHANE WOULD SEPARATE FROM WATER GIVING TWO SEPARATE ENVIRONMENTS, POTENTIALLY ALLOWING FOR INTERESTING CHEMISTRY AT THE POLAR-NONPOLAR BARRIER].

5 Conclusion

Titan presents a compelling setting for prebiotic chemistry. While likely entirely uninhabitable at its surface, its toxic tholins, once hydrolysed and in solution with cometary organics, could allow for fascinating chemistry. This study has examined impact cratering as a way to carry surface organics toward the ocean through the migration of brine pockets that hold high concentrations of varied compounds. These pockets offer a contained environment for complex chemistry before the products join the highly dilute ocean, if said ocean exists [Cite JPL]. Through producing an analytical impact model, the molecular inventory of a brine pocket formed by a 1.5 km cometary impactor was estimated and reduced to an analogue of 86 molecules, which has been simulated with GFN2-xTB MD across six independent replicas.

These simulations demonstrated that prebiotically significant chemistry does spontaneously occur in these environments. New carbon-nitrogen, carbon-phosphorus, and carbon-sulphur bonds form and consistently react with other organics, and glycine is mobilised into its glycyl fragment in every replica rather than being inert. Water, methane, ammonia, and the aromatic rings persist as a stable solvent [MAYBE NOT TRUE, CHANGE WITH FINAL DATA]. These outcomes are reproducible across six independent simulations. Due to the limitations brought on by the computational constraints, the data must be read as a screen for accessible chemistry rather than a measurement of rates or accurate concentrations. Within the imposed limits, an impact-generated Titan brine pocket is demonstrated to be able to support the first reactions towards prebiotic chemistry, encouraging further in-silica and in-vitro studies, as well as reaffirming that the refrozen remains of such brines are fascinating targets for future space exploration.

References [Reorder and add]

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Appendix 1

Table A1: Ideal composition of a simulation box calculated from the molecular inventory of the melt pool modelled in sections 2.1 and 2.2.

Species	Formula	Molecular abundance [%]	Total mass [kg]	Origin	Used in simulation [Y/N]	Source [PUT IN 124342]
Water	H_2O	84	4.86E12	Melt and cometary ice	Y	Durham et al., Rubin et al.
Carbon dioxide	CO_2	0.26	3.7E10	Cometary ice	Y	Rubin et al.
Carbon monoxide	CO	0.11	9.79E9	Cometary ice	Y	Rubin et al.
Molecular oxygen	O_2	0.080	8.28E9	Cometary ice	N	Rubin et al.
Hydrogen sulfide	H_2S	0.064	7.06E9	Cometary ice	Y	Rubin et al.
Phosphorus monoxide	PO	3.8E-4	5.82E7	Cometary ice	Y	Rubin et al.
Ethanol	C_2H_5OH	4.8E-5	7.23E6	Cometary organics	N	Schumann et al.
Ethylene glycol	$HOCH_2CH_2OH$	1.4E-5	2.75E6	Cometary organics	N	Schumann et al.
Methylamine	CH_3NH_2	0.40	4.02E10	Cometary organics and tholins	Y	Altwegg et al., and Imanaka et al.
Ethylamine	$C_2H_5NH_2$	1.2E-3	1.81E8	Cometary organics	N	Altwegg et al.
Hydrogen isocyanide	HNC	0.038	3.32E9	Cometary organics and tholins	Y	Altwegg et al., and Imanaka et al.
Formamide	NH_2CHO	0.014	1.99E9	Cometary organics	Y	Altwegg et al.
Formaldehyde	HCHO	0.27	2.62E10	Cometary organics	Y	Altwegg et al.
Thioformaldehyde	H_2CS	0.13	2.00E10	Cometary organics	Y	Altwegg et al.
Methanimine	CH_2NH	0.38	2.56E10	Cometary organics and tholins	Y	Altwegg et al., and Imanaka et al.
Ethene	C_2H_4	0.12	1.07E10	Cometary organics	N	Altwegg et al.

Methanol	CH_3OH	0.28	2.94E10	Cometary organics	Y	Altwegg et al.
Ethane	C_2H_6	0.49	4.70E10	Cometary organics and tholins	N	Altwegg et al., and Imanaka et al.
Nitrosoformaldehyde	HCONO	5.0E-3	9.82E8	Cometary organics	N	Altwegg et al.
Methane	CH_4	12	6.54E11	Melt and cometary organics	Y	Durham et al., and Altwegg et al.
Isocyanic acid	HNCO	8.7E-3	1.21E9	Cometary organics	Y	Altwegg et al.
Acetaldehyde	CH_3CO	0.084	1.21E10	Cometary organics	Y	Altwegg et al.
Formic acid	HCOOH	0.019	2.79E9	Cometary organics	Y	Altwegg et al.
Hydrogen cyanide	HCN	0.42	3.62E10	Cometary organics and tholins	Y	Altwegg et al., and Imanaka et al.
Ammonia	NH_3	2.5E-3	1.37E8	Cometary organics	Y	Altwegg et al.
Glycine	$C_2H_5NO_2$	3.7E-7	9.01E4	Cometary organics	Y	Gómez de Castro and de Isidro Gómez
Pyrrole	C_4H_5N	0.38	8.17E10	Tholins	Y	Imanaka et al.
Pyridine	C_5H_5N	0.38	9.63E10	Tholins	Y	Imanaka et al.
Niningerite (Fe)	FeS	0.053	1.50E10	Cometary minerals	N	Lisse et al.
Niningerite (Mg)	MgS	0.053	9.62E9	Cometary minerals	N	Lisse et al.
Forsterite	Mg_2SiO_4	0.080	3.65E10	Cometary minerals	N	Lisse et al.
Ferrosilite	$Fe_2Si_2O_6$	0.057	4.89E10	Cometary minerals	N	Lisse et al.
Amorphous Olivine	$MgFeSiO_4$	0.040	2.24E10	Cometary minerals	N	Lisse et al.